

Iterative Method:

An iterative method is a method in which a problem is solved by repeatedly executing a set of statements using loops such as for or while.

Algorithm:

Step 1: Start

Step 2: Read n from the user

Step 3: Set fact = 1

Step 4: Repeat for i = 1 to n

- fact = fact × i

Step 5: Display fact

Step 6: Stop

out put:

```
Enter a number: 2
Factorial = 2
Execution Time = 1.820800025598146e-05 seconds
Time Complexity = O(n)

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity: $O(n)$

Space Complexity: $O(1)$